TRIPOD checklist for prediction model development and validation

Completed against `manuscript/manuscript_expanded.md` (this study is a validation of a previously proposed calibration-to-accuracy relationship, not a development of a new model; items specific to model development are marked not applicable and the reason is given). Section and table references below follow the manuscript's current headings; where a checklist item is only partly addressed in the text, that gap is stated rather than papered over.

Section	Item	#	Checklist item	Reported on page/section
Title and abstract	Title	1	Identify as a validation study; specify the target population and outcome	Title; Abstract, Objective
	Abstract	2	Structured summary of objectives, design, setting, participants, predictor, outcome, statistical analysis, results, conclusions	Abstract
Introduction	Background	3a	Explain the medical context and rationale	Introduction, paragraphs 1-2
	Objectives	3b	State study objectives, including whether developing or validating	Introduction, final paragraph; Methods, Study Design and Reporting
Methods	Source of data	4a	Study design, data source	Methods, Study Design and Reporting (design); Methods, Data Source and Cohort (archive)

Section	Item	#	Checklist item	Reported on page/section
		4b	Study dates	Methods, Data Source and Cohort (archive version and digest; no recruitment dates, because this is a legacy public archive)
	Participants	5a	Setting and locations of centres	Not fully reported; the archive documentation does not specify recording sites or locations for the individual source studies (Methods, Data Source and Cohort)
		5b	Eligibility criteria	Methods, Data Source and Cohort
		5c	Treatments received, if relevant	Not applicable; this is a retrospective secondary analysis of archived P300-speller task-performance data with no clinical treatment or intervention (Methods, Ethics: “no new data collection, no participant contact, and no intervention”)
	Outcome	6a	Outcome definition and how/when assessed	Methods, Outcome

Section	Item	#	Checklist item	Reported on page/section
		6b	Blinding of outcome assessment	Not applicable; outcome is a deterministic reconstruction from archived event logs, not an assessor judgement
	Predictors	7a	Definition and measurement of predictors	Methods, Calibration Predictor; Methods, What the Calibration Score Is, and What It Is Not
		7b	Blinding of predictor assessment	Methods, Study Design and Reporting (calibration and outcome data are kept to separate protocol phases, not a demonstrated temporal precedence)
	Sample size	8	Explain how sample size was arrived at	Methods, Data Source and Cohort (all eligible archive studies used, not a powered sample)
	Missing data	9	Handling of missing predictor/outcome data	Methods, Calibration Predictor (epoch- and session-level inclusion thresholds); Methods, Outcome (feedback-phase eligibility); Results, Cohort; Table 1

Section	Item	#	Checklist item	Reported on page/section
	Statistical analysis methods	10a	Development: how predictors were handled in the analyses	Methods, Model Specification (calibration score standardised to the development-study mean and standard deviation and entered as a single continuous term)
		10b	Development: type of model, model-building procedures (including predictor selection), and method for internal validation	Methods, Model Specification (logistic regression of character-level correctness on the single, pre-defined calibration score; no predictor-selection procedure, because only one predictor was evaluated). This design has no internal-validation step distinct from the leave-one-cohort-out procedure itself; that procedure is the validation and is reported under 10c

Section	Item	#	Checklist item	Reported on page/section
		10c	Validation: how the predictions were calculated	Methods, Model Specification (predicted accuracy is the inverse logit of a + b(s-m)/d, using that fold's fitted coefficients, Supplementary Table S12); Methods, Validation Design (one source study withheld at a time)
		10d	Measures used to assess model performance	Methods, Statistical Analysis (mean absolute error against development-mean and held-out-cohort-mean benchmarks; root mean squared error; character-weighted Brier score and Brier skill score; calibration intercept and slope; character-weighted area under the curve)
		10e	Validation: any model updating	Not applicable; the fitted mapping from each development fold is applied as-is to its withheld cohort, not updated or recalibrated

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	Risk groups	11	How risk groups were created	Not applicable; outcome is a continuous accuracy proportion, not a risk category
	Development vs. validation	12	Differences between development and validation data	Methods, Study Design and Reporting (every cohort comes from one harmonised archive; the source study, not a separately assembled dataset, is the withheld unit); Discussion, Study Limitations (cohorts differ in speller matrix, stimulus paradigm, electrode type, and stopping rule)
Results	Participants	13a	Flow of participants, with reasons for exclusion	Results, Cohort; Table 1
		13b	Characteristics of participants	Results, Cohort (participant, session and selection counts); Table 2 (per-cohort composition, embedded under Results, Transportability)

Section	Item	#	Checklist item	Reported on page/section
		13c	Comparison of development vs. validation participants	Discussion, Study Limitations (cohort heterogeneity discussed; no separate person-level comparison table because cohorts, not held-out individuals, are the unit withheld)
	Model development	14a	Number of participants and outcome events in each analysis	Results, Cohort (271 participants, 410 sessions, 739 records, 19,611 analysed character selections; the outcome is a continuous per-record accuracy proportion rather than a discrete event count, see item 11)
		14b	Unadjusted association between each candidate predictor and outcome, if done	Results, Association Between Calibration-Derived Decoder Discriminability and Online Accuracy
	Model specification	15a	Full prediction model (all coefficients)	Supplementary Table S12 (fitted coefficients of every development fold)

Section	Item	#	Checklist item	Reported on page/section
		15b	Explanation of how to use the model	Methods, Model Specification (closed-form estimate, a + b(s-m)/d); Supplementary Table S12 (worked numeric example)
	Model performance	16	Performance measures with confidence intervals	Results, Estimation Error and Its Reference Points (pooled measures); Results, Transportability (per-cohort heterogeneity, including Brier skill; Table 2; Table S2; Figures 1-4)
	Model updating	17	Details of model updating	Not applicable; see item 10e
Discussion	Limitations	18	Study limitations, sources of bias	Discussion, Study Limitations
	Interpretation	19a	Interpretation for validation studies, with reference to performance in the development data	Discussion, paragraph 6 (fold-coefficient stability: development-fold intercept and slope coefficients vary by only 2.5% and 3.3% across folds, against 49.4% for the held-out cohort-specific validated slopes in Table S2)
		19b	Overall interpretation, implications for practice	Discussion, Conclusion

Section	Item	#	Checklist item	Reported on page/section
	Implications	20	Potential clinical use, implications for future research	Discussion, Conclusion; Abstract, Significance
Other information	Supplementary information	21	Availability of supplementary resources	Data and Code Availability (data and code); the Supplementary Information document (`supplementary/supplement_ex` and this checklist (`supplementary/tripod_checkl` are submitted as separate companion files alongside the manuscript, not referenced by filename in the main text
	Funding	22	Source of funding	Acknowledgements